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Seat No : _____

No. of Printed Pages : 2

**Sardar Patel University
External Examination**

M.Sc. (Information Technology) Semester – I

PS01CINT24: Operating System Concepts

23rd November, 2019, Saturday

Time: 10:00 AM to 01:00 PM

Max. Marks: 70

Q-1 Choose the most appropriate option for each question.

[08]

- (i) Which of the following is not a component of operating system?
 - a) System Program
 - b) Application Program
 - c) Loader
 - d) Linker
- (ii) Full form of DMA is _____.
 - a) Distributed Memory Access
 - b) Direct Memory Access
 - c) Distributed Memory Allocation
 - d) Direct Memory Allocation
- (iii) _____ Model maps many user level threads to one kernel thread.
 - a) Many to One
 - b) One to Many
 - c) Many to Many
 - d) All of these
- (iv) The operating system starts in _____ mode.
 - a) User
 - b) Kernel
 - c) Supervisor
 - d) Both (b) and (c)
- (v) Which of the following algorithm is not a stack algorithm?
 - a) FIFO
 - b) LRU
 - c) Optimal
 - d) None of these
- (vi) Which of the following structure is used in segmentation?
 - a) Page Table
 - b) Segment Table
 - c) TLB
 - d) None of these
- (vii) Which of the following is NOT a type of file?
 - a) .jpg
 - b) .org
 - c) .jpeg
 - d) .doc
- (viii) Which of the following algorithm is used for deadlock avoidance?
 - a) Resource Allocation
 - b) Banker's
 - c) Peterson's
 - d) None of these

Q-2 Answer ANY SEVEN from the following

[14]

- (i) List roles of operating systems.
- (ii) Differentiate: shared memory v/s message passing systems
- (iii) What is API?
- (iv) What do you mean by non preemptive CPU Scheduling algorithm?
- (v) What is swapping?
- (vi) Briefly explain execution time address binding.
- (vii) What is page fault?
- (viii) What are major benefits of using RAID?
- (ix) Briefly explain any two file operations.

(1)

(PTO)

Q-3

A. What is operating system? Explain services provided by operating systems to the users. [06]

B. Explain (i) Privileged Instructions and (ii) System Call [06]

OR

B. Write a note on virtual machines.

Q-4

A. Draw and explain PCB. [06]

B. What is deadlock? Explain necessary conditions for deadlock. [06]

OR

B. Explain FCFS CPU scheduling algorithm by giving suitable example of your choice. Draw Gantt chart and calculate average waiting time for the same.

Q-5

A. Write a note on segmentation. [06]

B. What is page table? List the possible structures of page table and explain any one of them in detail.. [06]

OR

B. Explain LRU page replacement algorithm by considering following reference string:

7,0,1,2,0,3,0,4,2,3,0,3,2,1,2,0,1,7,0,1

Assume that each process is given three frames in the memory.

Calculate number of page faults for the same.

Q-6

A. Define the terms "Seek Time", "Disk Bandwidth", and "Rotational Latency" in context of hard disk. Draw the structural diagram of Hard Disk Drive. [06]

B. Explain (i) SSTF and (ii) LOOK disk scheduling algorithms with examples. [06]

OR

B. What do you mean by different levels of RAID? Explain any two levels of RAID.

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(2)